

Vapor-Liquid Equilibrium of MTBE-Methanol, MTBE-Methanol-Calcium Chloride and MTBE-Methanol-Lithium Chloride Mixtures

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Isobaric vapor-liquid equilibria of methyl *tert*-butyl ether (MTBE)-methanol, MTBE-methanol-calcium chloride, and MTBE-methanol-lithium chloride mixtures were measured at 93.57 kPa using a Malanowski equilibrium still which circulates both the vapor and liquid phases. The experimental results of the salt-free system showed that MTBE-methanol forms a minimum boiling azeotrope of 69.80 mol.-% MTBE at 93.57 kPa and 322.67 K. The experimental results of the salt-free system were predicted using the original UNIFAC and the UNIFAC-Dortmund methods, where the vapor-phase compositions were predicted with root-mean-square deviations (RMSD) of 0.0046 and 0.0091, respectively, and the bubble point temperatures were predicted with RMSD of 0.36 and 0.71 K, respectively. The Wilson and the nonrandom two-liquid (NRTL) models could satisfactorily correlate the experimental data of the salt-free mixtures with RMSD in the vapor-phase compositions calculation of 0.0047 and 0.0059, respectively and in bubble-point temperatures calculation of 0.20 and 0.40 K, respectively. Addition of either calcium chloride or lithium chloride to the MTBE-methanol mixtures resulted in salting-out the MTBE and shifting the azeotropic point to a vapor composition of 76.5 mol.-% and 93.0 mol.-% MTBE, respectively. Both the Tan-Wilson and Tan-NRTL models could satisfactorily predict the bubble-point temperatures of the MTBE-methanol-calcium chloride mixtures with RMSD of 0.143 and 0.130 K, respectively, and the vapor phase compositions with RMSD of 0.0055 and 0.0041, respectively.

1 Introduction

Methyl *tert*-butyl ether (MTBE) is an oxygenated petrochemical with excellent antiknock properties. The use of MTBE pure or mixed with alcohols has been recommended as a high-octane blending agent for motor gasoline [1,2]. Thus, MTBE is rapidly emerging as one of the most successful octane booster for lead-free or low-leaded gasoline. On the other hand, high-purity MTBE is also increasingly valued as a solvent and a chemical reactive compound. MTBE has advantages over other ethers in that it minimizes handling and storage problems associated with peroxide formation and exhibits excellent oxidative stability, preventing peroxide formation without the need to add inhibitors [3].

MTBE is produced on a commercial scale using reactive distillation by heterogeneous catalytic reaction of isobutylene and methanol in the liquid phase over a strong acidic ion-exchange resin catalyst. To achieve higher conversion, excess methanol can be used [4]. However, this excess methanol may introduce a purification problem as this system exhibits a minimum-boiling azeotrope at about 70 mol.-% MTBE and 101.3 kPa [5]. Coto *et al.* [6] studied the VLE of MTBE-methanol mixtures over a wide range of temperature and found that the azeotropic point of this system occurs at about 70 mol.-% MTBE at 324.3 K and about 80 mol.-% MTBE at 298.15 K. Thus, conventional distillation cannot be used to

separate this system for higher MTBE concentrations. Other separation process alternatives have been suggested, such as pervaporation [7] and hybrid distillation-pervaporation process [8]. Loras *et al.* [5] studied the azeotropic behavior of the MTBE-methanol mixtures at different pressures as another separation process alternative. Velasco *et al.* [9] studied the effect of inorganic salts on the VLE of this azeotropic system and they showed that salt has a favorable effect on the separation of the MTBE-methanol mixtures.

Addition of inorganic salts to mixed-solvent mixtures can alter the phase equilibrium of these mixtures. This alteration is due to the interaction of the added salts with the components of these mixtures. In most cases, the molecules of the most polar component in the mixture will be preferentially attracted by the electrostatic field of the ions constituting the salt resulting in enriching the vapor phase with the less polar component in which the salt is less soluble. The effect of salt addition on VLE has been studied experimentally and theoretically by many investigators. Furter [10], Furter and Cook [11], Ciparis [12], Ohe [13] and Kumar [14] made excellent reviews on the principles and technical details related to the salt effect in addition to collected experimental data of VLE of systems containing salts.

The objectives of this work are to measure the VLE of MTBE-methanol mixtures at atmospheric pressure of 93.57 kPa and to study the effect of salt addition of either calcium chloride or lithium chloride on the VLE of this system. The measured VLE salt-free data are to be compared with predicted data by two activity-coefficient methods; the original UNIFAC and the UNIFAC-Dortmund methods. Further, they will be correlated with the Wilson and the NRTL models. Finally, the effect of salt addition is to be analyzed using the Tan and coworkers approach.

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2 Experimental

2.1 Chemicals

MTBE (purity: 99.8 %, supplier: Acros) and methanol (purity: 99.5 %, supplier: GCC) were used in this study without further purification. Purity of these chemicals was checked by reproducing their boiling points at the operating pressure. The salts used were calcium chloride (purity: +99 %, supplier: Analr) and lithium chloride (purity: +99 %, supplier: Acros).

2.2 Procedure

The VLE data of the studied systems were measured using a modified Malanowski equilibrium still which circulates both the vapor and liquid phases through a Cottrell pump. A detailed description of this still can be found elsewhere [15]. The pressure in the still was measured using a mercury column barometer with an accuracy of ± 0.10 Pa. Steady state was reached after 30 min, where the temperature was measured using a calibrated mercury-in-glass Beckman thermometer with a resolution of ± 0.01 K. For the salt experiments, mixtures were prepared using methanol saturated with the salt.

2.3 Sample Analysis

Samples of liquid and condensed vapor were withdrawn and analyzed with a precision better than 0.0002 mole fraction, using a gas chromatograph (Varian 3400) with a flame ionization detector (FID) and a column with the following specifications: type: 10 % dodecyl phthalate Chromosorb W/HP 80/100, length: 4.0 m, and diameter: 0.3175 cm. The column temperature was 413.15 K. Nitrogen with a flow rate of 20 ml/min was used as the carrier gas.

In the case of lithium chloride addition, immiscible regions were obtained. In these regions the samples of both liquid and vapor phases were collected using the method of Lin *et al.* [16]. However, the tabulated data were those corresponding to the overall liquid composition which were assumed to be similar to those of the feed.

3 Results and Discussion

3.1 VLE of MTBE-Methanol Salt-Free Mixtures

The experimental isobaric VLE data for the MTBE-methanol mixtures at 93.57 kPa are tabulated in Tab. 1 and presented in Fig. 1, as a T - x - y diagram. These data show that the system MTBE-methanol at 93.57 kPa has an azeotropic point at 69.80 mol.-% MTBE and 322.67 K. A comparison between the experimental data of this study and other studies in terms of azeotropic point is shown in Tab. 2.

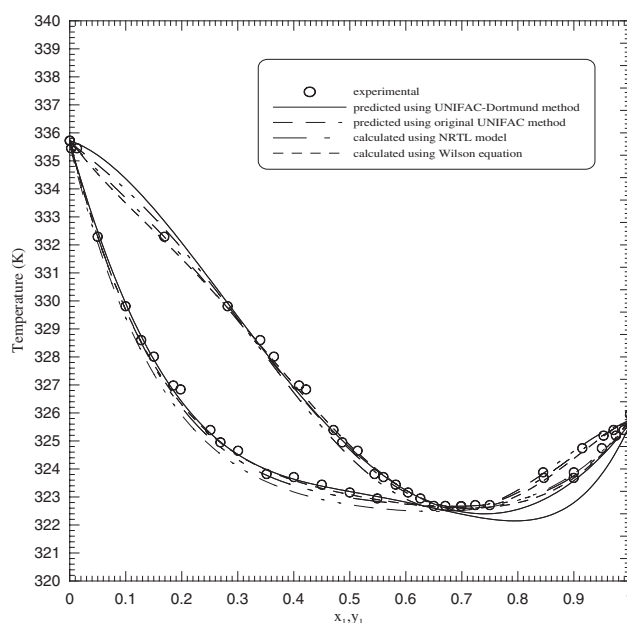


Figure 1. Isobaric x - y - T data of MTBE-methanol mixtures at 93.57 kPa.

Table 1. Isobaric VLE data for MTBE(1)-methanol(2)-salt systems at 93.57 kPa.

Antoine constants [5]:

$$\begin{array}{lll} A_1 = 6.7359 & B_1 = 1032.99 & C_1 = 213.27 \\ A_2 = 7.8975 & B_2 = 1474.08 & C_2 = 229.27 \end{array}$$

Wilson parameters:

$$\Delta\lambda_{12} = -10479, \Delta\lambda_{21} = 14282 \text{ (J/mol)}$$

NRTL parameters:

$$\Delta g_{12} = 2025.3, \Delta g_{21} = 2133.5 \text{ (J/mol)}, \alpha = 0.60$$

Salt-free	Saturated with CaCl ₂					Saturated with LiCl			
	$A_1^{\text{f}} = 1.00$					$A_1^{\text{f}} = 1.00$			
	$A_2^{\text{f}} = 1.07$					$A_2^{\text{f}} = 1.35$			
T(K)	x_1	y_1	T(K)	x_1	y_1	T(K)	x_1	y_1	
335.72	0.0000	0.0000	337.45	0.0000	0.0000	343.55	0.0000	0.0000	
335.45	0.0025	0.0130	334.96	0.0356	0.1526	334.95	0.0800	0.7100	
332.29	0.0499	0.1689	333.59	0.0548	0.2126	333.04	0.1100	0.9100	
329.81	0.0997	0.2819	330.04	0.1247	0.3606	325.75	0.3700	0.9140	
328.60	0.1275	0.3401	327.71	0.1985	0.4540	324.24	0.5300	0.9170	
328.01	0.1501	0.3645	326.54	0.2515	0.5007	323.41	0.7230	0.9170	
326.98	0.1850	0.4094	325.39	0.3120	0.5400	323.41	0.8300	0.9170	
326.84	0.1979	0.4219	324.61	0.3929	0.5855	324.42	0.9480	0.9370	
325.39	0.2511	0.4709	324.08	0.4370	0.6080	325.98	1.0000	1.0000	
324.95	0.2691	0.4865	324.02	0.4970	0.6180				
324.65	0.3005	0.5139	323.45	0.5402	0.6378				
323.82	0.3514	0.5441	323.23	0.6783	0.7000				
323.71	0.4001	0.5601	323.13	0.7443	0.7495				
323.43	0.4500	0.5820	323.17	0.7647	0.7651				
323.15	0.5000	0.6040	323.43	0.8349	0.7840				
322.94	0.5490	0.6260	323.79	0.8796	0.8202				
322.68	0.6500	0.6705	324.52	0.9315	0.8764				
322.67	0.6989	0.6986	325.95	1.0000	1.0000				
322.71	0.7499	0.7240							
323.68	0.8998	0.8464							
323.88	0.9001	0.8452							
324.74	0.9501	0.9151							
325.19	0.9751	0.9531							
325.39	0.9891	0.9709							
325.95	1.0000	1.0000							

Table 2. Azeotropic points for MTBE-methanol system.

P (kPa)	T(K)	x_1	Ref.
36.24	298.15	0.7919	[6]
32.84	296.15	0.7994	[16]
35.27	298.15	0.8040	[9]
39.36	300.15	0.7865	[17]
43.75	303.15	0.7700	[18]
44.55	303.15	0.7697	[6]
53.33	307.64	0.7690	[19]
54.43	308.15	0.7520	[6]
80.44	318.15	0.7103	[6]
162.65	338.15	0.6306	[6]
79.63	318.15	0.7270	[20]
93.57	322.67	0.6980	this work
94.00	322.48	0.6860	[5]
100.50	323.95	0.6994	[17]

The thermodynamic consistency of the experimental data of this study was examined by the infinite-dilution test of Kojima *et al.* [21] as modified by Jackson and Wilsak [22] and the integral test as modified by Herington [23]. The infinite dilution test of Kojima-Jackson-Wilsak involves the calculation of the excess Gibbs energy from the experimental data and then extrapolation to the infinite dilution. A comparison is then to be made between these extrapolated values and those values obtained by extrapolating the $\ln \gamma_1$ and $\ln \gamma_2$ curves to infinite dilution. According to this test, the VLE data are consistent if the values agree within 30 %. Upon applying this test to our experimental data it suggests that the experimental data of this study can be assumed to be thermodynamically consistent, where the error for dilute $\ln \gamma_1$ and $\ln \gamma_2$ are found to be 6.5 and 8.3 %, respectively. According to the Herington test, consistency of VLE data requires $|D - J| < 10$, where¹⁾

$$D = \frac{100 \left| \int_0^1 \ln \frac{\gamma_1}{\gamma_2} dx_1 \right|}{\Sigma}$$

Σ is the sum of the absolute values of the areas under the curve

$$\ln \frac{\gamma_1}{\gamma_2} \text{ vs. } x_1, \quad J = \frac{150|\phi|}{T_{\min}}$$

ϕ is the maximum boiling point of the system, and $T_{\min}$ (K) is the lowest measured boiling point of the system. Applying this test to our data results in $|D - J| = 4.87$, which suggests that the experimental data can be assumed to be consistent.

1) List of symbols at the end of the paper.

3.2. Prediction and Correlation of the Salt-Free Experimental Data

The experimental results of the salt-free system have been compared with those predicted by the original UNIFAC [24] and the UNIFAC-Dortmund [25] group contribution methods, and the results are plotted in Fig. 1. Both methods gave good results in predicting either the bubble-point temperatures or the vapor-phase compositions where the vapor-phase compositions were predicted with root-mean-square deviations (RMSD) of 0.0046 and 0.0091, respectively, and the bubble point temperatures were predicted with RMSD of 0.36 and 0.71 K, respectively.

The experimental data were further analyzed by correlating them using the Wilson [26] and the NRTL [27] models for liquid-phase activity coefficients. The interaction parameters of the two models were estimated by minimizing the following objective function

$$O.F. = \sum_{j=1}^n \left[(\gamma_{1,j,\text{exp}} - \gamma_{1,j,\text{cal}})^2 + (\gamma_{2,j,\text{exp}} - \gamma_{2,j,\text{cal}})^2 \right] \quad (1)$$

where n is the number of the experimental data points. This objective function was minimized using the Gradient Search Optimization method. The resulted optimum parameters, shown in Tab. 1, are then used to calculate the bubble-point temperatures and the vapor-phase compositions corresponding to the measured pressure and liquid-phase compositions and the results are plotted in Fig. 1. As can be noticed from Fig. 1, both models can be adequately used to calculate the VLE of the MTBE-methanol mixtures, with the Wilson model giving better agreement with the experimental data with 0.20 K RMSD in bubble-point temperatures calculation and 0.0047 RMSD in vapor-phase compositions calculation, while calculations using the NRTL model result in 0.40 K RMSD in bubble-point temperatures calculation and 0.0059 RMSD in vapor-phase compositions calculation.

3.3 Effect of Salt Addition on the VLE of MTBE-Methanol Mixtures

The experimental data for MTBE-methanol-calcium chloride and MTBE-methanol-lithium chloride mixtures are shown in Tab. 1 and presented in Fig. 2. These results indicate that both salts resulted in altering the VLE of the MTBE-methanol mixtures and salting-out the MTBE. As can be noticed from these results, the effect of lithium chloride was much more pronounced where it resulted in shifting the azeotropic point of the system from 69.8 mol.-% MTBE to 93.00 mol% MTBE, compared to that of 75.5 mol.-% MTBE in the presence of calcium chloride. Moreover, a vapor composition of 91.0 mol.-% MTBE corresponding to a liquid composition of 11.0 mol.-% MTBE was obtained. Abu Al-Rub and Datta [28] and Kumar and Rajendran [29] attributed such strong effect of the salt to the formation of possible methanol-lithium ions complex due to stronger intermolecu-

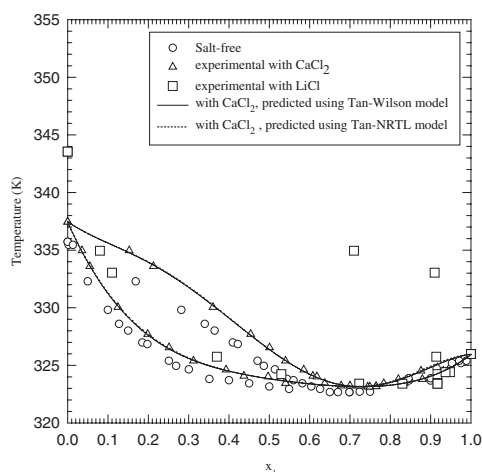


Figure 2. Isobaric x - y - T data of MTBE-methanol-salt mixtures at 93.57 kPa.

lar forces between them more than either MTBE-methanol or MTBE-lithium ions. These results can also be explained using the Tan and coworkers approach based on the solute-solvent interaction parameter, A_i^s , which is defined by [30–36]

$$A_i^s \equiv \frac{p_i^s}{P} \text{ evaluated at } T_{\text{salt},i} \quad (2)$$

where $T_{\text{salt},i}$ is the bubble point of the solvent component i saturated with dissolved solute or having the same solute concentration at the given system according to whether the system is saturated or unsaturated but containing a constant solute concentration over the whole mole fraction range. The values of the salt-solvent interaction parameters were calculated for MTBE and methanol in the presence of either of the salts and the calculated values are listed in Tab. 1. An effective measure of the solute effect on the solvent mixture can be obtained by using the ratio of the solute-solvent interaction parameters for both solvents, A_{21}^s , where

$$A_{21}^s \equiv \frac{A_2^s}{A_1^s} \quad (3)$$

Values of A_{21}^s greater than one indicate salting-out effect of the lighter component. The values of A_{21}^s for our system in the presence of calcium chloride and lithium chloride were found to be 1.07 and 1.35, respectively, which indicates salting-out of the MTBE. The value of A_{21}^s in the presence of lithium chloride is considered very large compared to unity, which means a large and considerable effect of the salt on the binary system. This is in accordance with the experimental results.

Next, the Tan-Wilson [31] and Tan-NRTL [36] models are used to predict the VLE of MTBE-methanol-calcium chloride mixtures. These models are based on using the solvent-solvent interaction parameters for the salt-free system and the solute-solvent interaction parameters, A_i^s , to predict the VLE of mixtures in the presence of salts. The predicted VLE data using the interaction parameters listed in Tab. 1 are shown in Fig. 2. The RMSD in bubble-point temperatures prediction using these models (0.143 K and 0.13 K, respectively) and

vapor-phase compositions prediction (0.0055 and 0.0041, respectively) indicate that the prediction of VLE of MTBE-methanol-calcium chloride using these models compared well with the experimental data. Due to the presence of immiscible regions in the presence of lithium chloride, the Tan-Wilson equation failed to calculate the VLE of MTBE-methanol-lithium chloride mixtures.

4 Conclusions

The VLE data obtained using the Malanowski still for the binary MTBE-methanol system at 93.57 kPa were shown to be thermodynamically consistent. The experimental results showed that this system exhibits an azeotropic point at 69.80 mol.-% MTBE and 322.67 K. The original UNIFAC method could satisfactorily predict both the bubble-point temperatures and the vapor-phase compositions of the MTBE-methanol mixtures. The Wilson and the NRTL models were adequate for correlating the experimental VLE data obtained in this study. The addition of either calcium chloride or lithium chloride to the MTBE-methanol mixtures could result in an alteration in the VLE of this system and could shift the azeotropic point to a composition of 93.0 mol.-% MTBE in the presence of lithium chloride. Both the Tan-Wilson and Tan-NRTL models have successfully predicted the effect of adding calcium chloride on the VLE data of MTBE-methanol mixtures.

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Symbols used

A, B, C	[–]	Antoine parameters
A_i^s	[–]	solute-solvent interaction parameter between solute (s) and solvent (i)
A_{21}^s	[–]	ratio of the solute-solvent interaction parameters of both solvents
D, J	[–]	terms in Herington test
T	[K]	temperature
$T_{\text{salt},i}$	[K]	the bubble point of the solvent component i saturated with dissolved solute (K)
x	[–]	salt-free mole fraction in the liquid phase
y	[–]	salt-free mole fraction in the vapor phase

Greek symbols

φ	[–]	the maximum boiling point of the system (K)
γ_i	[–]	activity coefficient of the i th component

Subscripts

1	MTBE
2	methanol
i	i th component

Superscript

s	solute
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Abbreviations

RMSD	root-mean-square deviations
VLE	vapor-liquid equilibrium

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